Triadic Interaction Geometry and the Breakdown of Pairwise Composition in Event Density

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Abstract
Event Density (ED) has now matured to the point where its architectural laws can be tested beyond the bilateral setting. ED-Arch-12 established a complete two-core interaction law, raising the central question of compositionality: does the bilateral law extend to three-core systems, or do new many-body mechanisms emerge? ED-Arch-13 answers this question through a sequence of controlled triadic experiments, including intruder geometries, symmetric three-core configurations, and a systematic sweep over initial separations and velocities.

Across all physically valid regimes, the two-core law fails to compose. Instead, triads exhibit new interaction modes: blocking by a bound pair, early field-capture of an intruder, overlap-driven cascade collapse, and spontaneous symmetry breaking in perfectly symmetric configurations. No stable or metastable three-core structure is found at the tested parameters; triads either decay before interacting or collapse rapidly once they do. From these results, we derive the Triadic Interaction Laws, which define a structured four-regime phase diagram and establish that ED's many-body dynamics are not reducible to pairwise terms.

ED-Arch-13 closes the bilateral era of ED architecture. It demonstrates that triadic interactions introduce qualitatively new mechanisms that require their own laws, setting the stage for ED-Arch-14, where dissipation and amplitude will be tuned to probe whether triadic stability can ever emerge.

1. Introduction
Event Density (ED) has now reached the point where its architectural laws can be tested beyond the pairwise setting. ED-Arch-12 established the first complete two-core interaction law: a clean MERGE/BIND/SCATTER classification with a quantitative threshold and reproducible kinematic predictions. That result raised an immediate architectural question: does the two-core law compose? If ED is to support many-body structure, then the bilateral law must either (a) extend cleanly to triads or (b) reveal new interaction regimes that only appear when three cores share a field.

ED-Arch-13 answers that question decisively. When a third core is introduced, the two-core law does not compose. Instead, the shared field of a bound pair generates new absorption channels, new collapse mechanisms, and spontaneous symmetry breaking that have no analogue in the bilateral system. Across a wide range of geometries, velocities, and separations, triads either decay before interacting or collapse rapidly once they do. No stable three-core configuration is found at the tested parameters.

This paper documents the full experimental arc that leads to the Triadic Interaction Laws: a set of architectural principles governing how three cores behave in ED. These laws mark the first departure from pairwise compositionality and establish the foundation for ED-Arch-14, where dissipation and amplitude will be tuned to probe whether triadic stability can ever emerge.

2. Methods

All experiments use the standard Scenario H shared-field configuration. The field evolves under the same convolution kernels, decay rate $\alpha = 0.001$, and update rules used in ED-Arch-12. Cores are initialized as Gaussian density peaks with identical amplitude unless otherwise stated. Centroids are tracked using the same blob-detection pipeline and the same MERGE threshold (≈ 23–24 px) derived in the two-core study.

Three experimental geometries are used:

1. Intruder Triads
A stable A–B pair is initialized in the BIND regime ($v = 0.20$, $b = 20$). A third core C approaches from above at controlled distances and velocities. This geometry tests whether the A–C MERGE predicted by the two-core law still occurs when B is present.

2. Symmetric Triads
Three identical cores are placed at the vertices of an equilateral triangle and directed toward the centroid. This geometry removes any privileged pair and tests whether ED supports a stable three-core configuration or whether symmetry breaks spontaneously.

3. R × v Sweep
The symmetric triad is repeated across a grid of circumradii ($R = 30, 40$) and inward speeds ($v = 0.05$–0.20) to search for long-lived triads and identify the boundary between natural decay and interaction-driven collapse. Larger R values (50, 60) are included for completeness but excluded from physical interpretation due to domain-wrapping artifacts.

For each run, the following quantities are recorded:

- core count over time
- MERGE events and their timestamps
- minimum pairwise separations $D_{\min}$
- whether $D_{\min}$ crosses the MERGE threshold
- centroid trajectories
- symmetry metrics (pairwise distance differences)
- final configuration (3, 2+1, 1, or 0 cores)

A configuration is considered "long-lived" if all three cores remain distinct for at least 2000 steps.

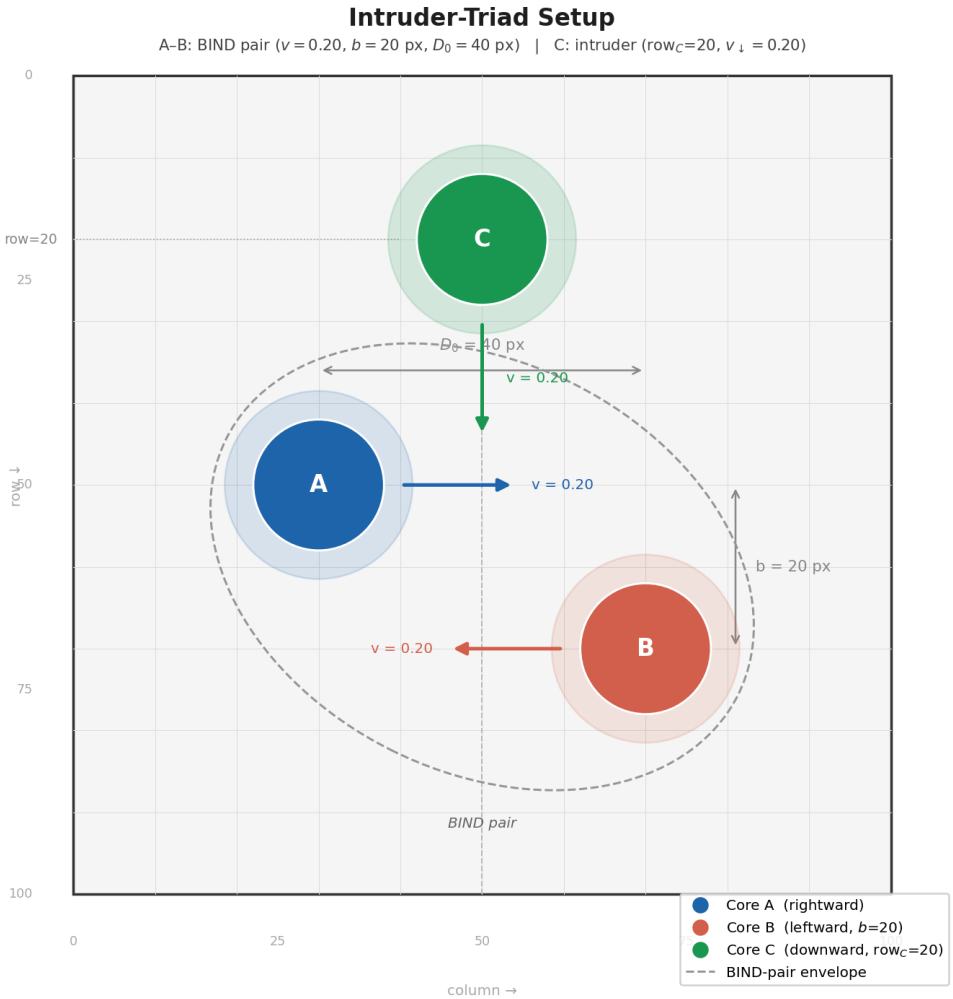
3. Experiment Series 1 — Intruder Triads

The first set of experiments probes the simplest triadic extension of the two-core law: a stable A–B pair with a third core C approaching from above. In isolation, the A–C pair would MERGE at predictable times depending on their initial separation. The question is whether this MERGE still occurs when B is present.

3.1 Slow Triad (400 steps)

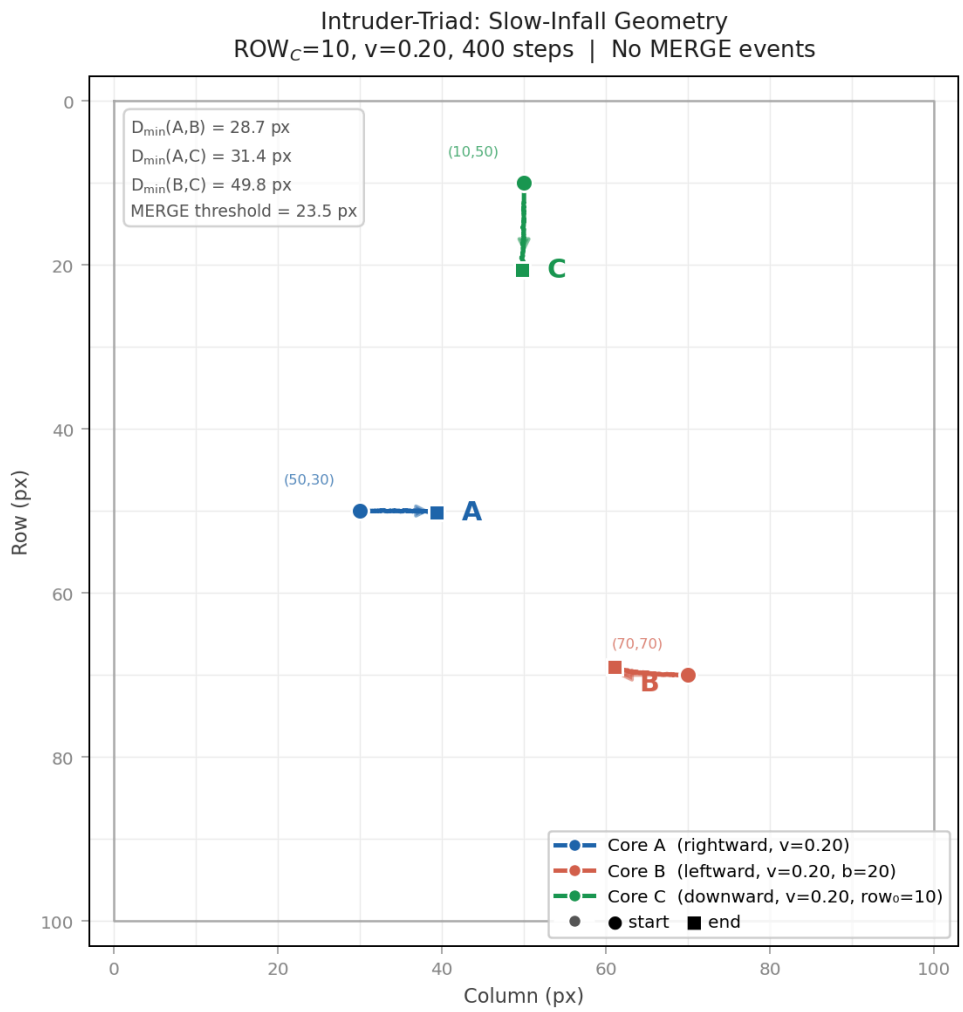
With C starting far above A, all three cores behave as independent slow scatterers. Pairwise distances remain large, no thresholds are crossed, and the two-core law appears to compose additively. This run serves as a baseline.

Figure 1: Intruder Triad Geometry and Experimental Setup



Schematic of the intruder-triad configuration used in Experiment Series 1. A and B form a bound pair in the BIND regime (v = 0.20, b = 20), while C approaches from above at controlled initial separations. All cores are initialized as identical Gaussian density peaks under Scenario H. This geometry tests whether the two-core MERGE law from ED-Arch-12 composes in the presence of a third core.

Figure 2: Slow Triad (400 Steps) — Independent Scatterers

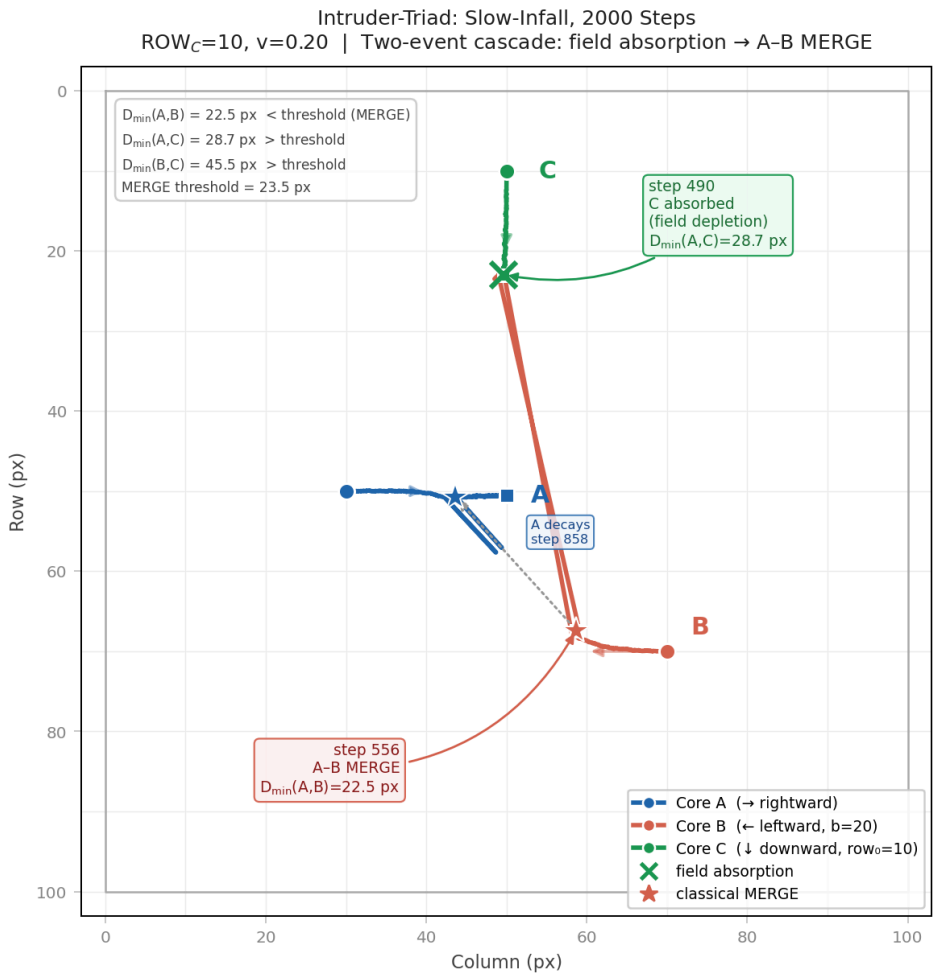


Centroid trajectories for the slow intruder triad. All three cores remain far apart, no MERGE thresholds are approached, and the system behaves as three independent scatterers. This run establishes the baseline regime where triadic effects are absent.

3.2 Long-Window Triad (2000 steps)

Extending the same geometry reveals the first triadic effect. Core C disappears at step 490 without ever reaching the MERGE threshold with A. This is not a classical MERGE; it is a field-absorption event driven by the shared A–B field. Later, A and B merge classically, and the resulting core decays. The triad does not follow a simple 3 → 2 → 1 ladder; instead, it exhibits a mixed mechanism: early triadic absorption followed by bilateral MERGE.

Figure 3: Long-Window Triad (2000 Steps) — Early Field Absorption

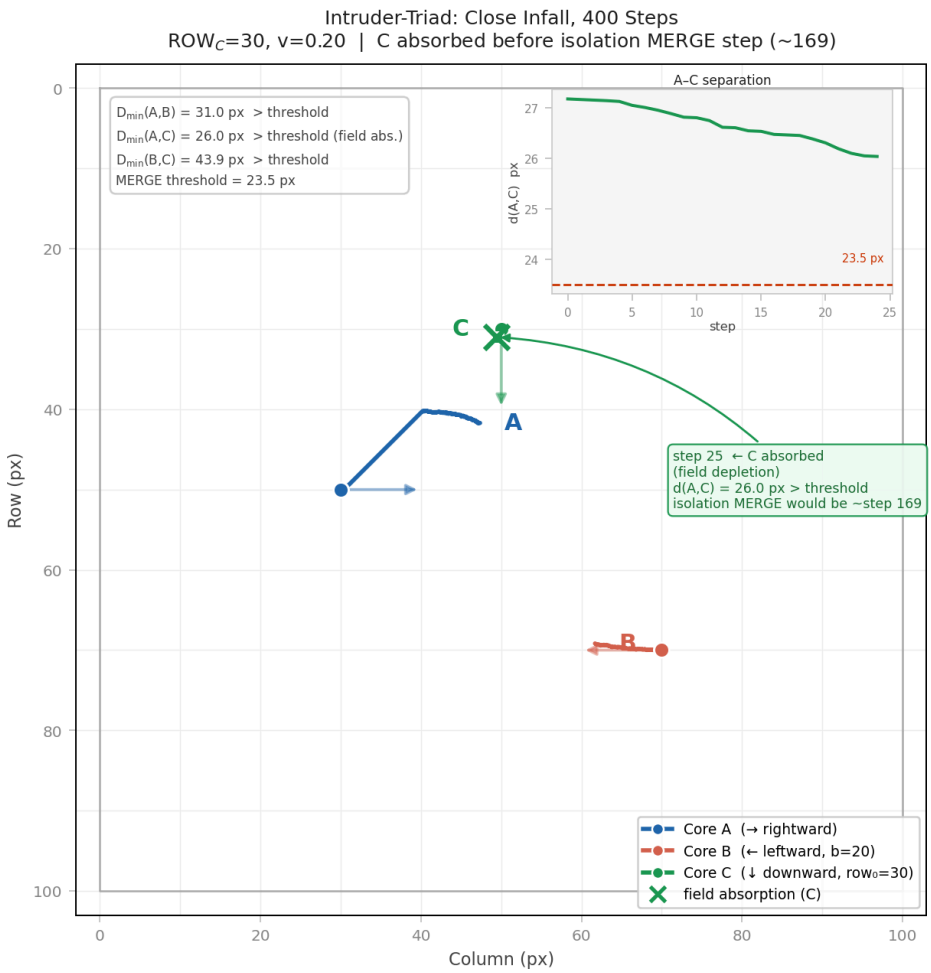


Core C is absorbed at step ~490 without crossing the MERGE threshold with A. The A–B pair later undergoes a classical MERGE. This run provides the first evidence that the two-core law does not compose: the shared A–B field dissolves C before any bilateral interaction can form.

3.3 Fast Intruder

When C is moved closer so that A–C would MERGE in isolation within ~170 steps, the triad suppresses this MERGE entirely. C is absorbed at step 25, with $D_{min}(A,C)$ still above threshold. The A–B pair acts as a dominant field attractor that dissolves C before any bilateral interaction can form. This is the first clear demonstration that the two-core law does not compose.

Figure 4: Fast Intruder — Suppression of the A–C MERGE



Despite starting in a geometry where A–C would MERGE in isolation within ~170 steps, C is absorbed at step 25 with D_min(A,C) still above the MERGE threshold. The A–B shared field blocks the bilateral MERGE channel, demonstrating the triadic blocking mechanism.

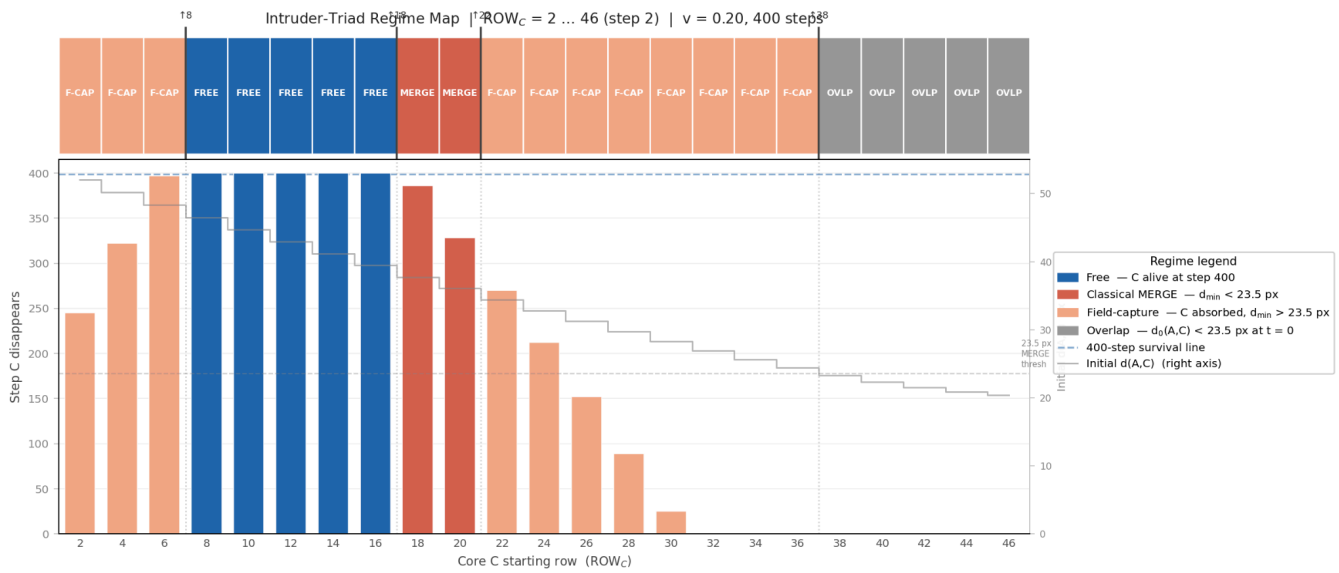
3.4 Intruder Distance Sweep

A systematic sweep of C's starting row reveals four distinct regimes:

- Free Regime — C too distant; no interaction within 400 steps.
- Classical MERGE Regime — C approaches A over hundreds of steps and crosses the MERGE threshold.
- Early Field-Capture Regime — C starts inside the A–B field and is absorbed before any kinematic approach.
- Overlap / Cascade Regime — C begins inside A's core; immediate absorption triggers a cascade.

The boundary between classical MERGE and early field-capture is a mechanism flip, not a smooth transition. This structured frontier is the first component of the Triadic Interaction Laws.

Figure 5: Intruder Distance Sweep — Four Triadic Regimes



Outcome map for C's initial row position. The sweep reveals four distinct regimes: (1) free motion, (2) classical MERGE, (3) early field-capture, and (4) overlap/cascade collapse. The boundary between classical MERGE and early capture is a mechanism flip, not a smooth threshold. This structured frontier defines the first triadic phase diagram in ED.

4. Experiment Series 2 — Symmetric Triads

The second experimental series removes the privileged A–B pair entirely. Instead of asking how a third core perturbs a bound pair, we ask whether ED can sustain a fully symmetric three-core configuration. If the two-core law were compositional, a symmetric triad might form a stable orbit or at least a long-lived metastable structure. If not, the triad should collapse or decay in a way that reveals new many-body mechanisms.

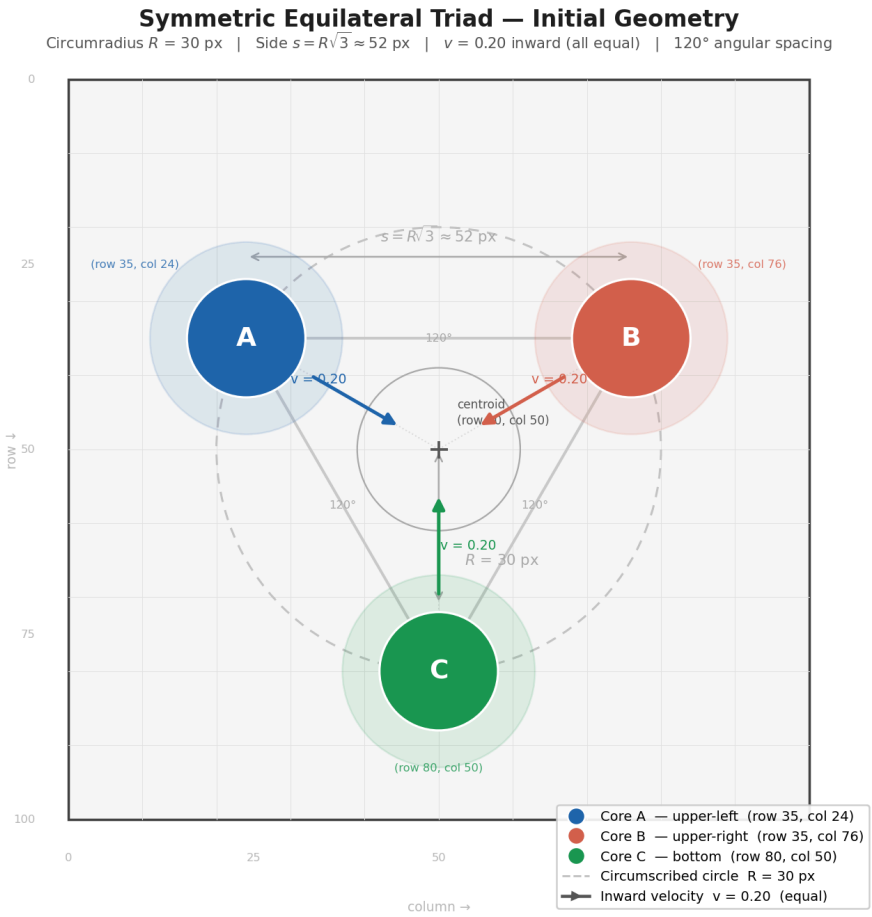
4.1 Symmetric Triad (400 steps)

Three identical cores are placed at the vertices of an equilateral triangle of circumradius $R = 30$ px, centered in the domain. All three velocities point toward the centroid with equal magnitude $v = 0.20$. This is the cleanest possible test of triadic symmetry.

For the first 400 steps, the symmetry is preserved to within numerical precision. All pairwise distances remain equal to within ~0.3 px, and no pair approaches the MERGE threshold. The cores have moved only ~8 px of their 30 px journey toward the centroid. No interactions occur on this timescale.

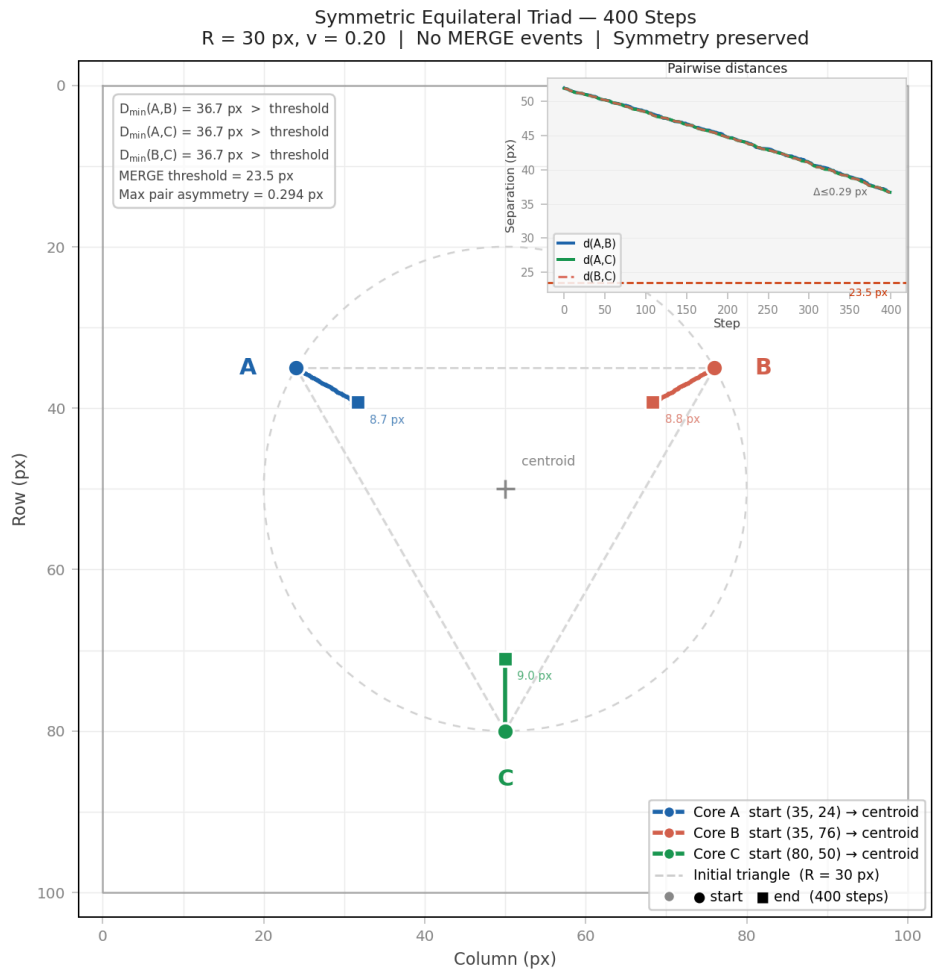
This run establishes the symmetric infall phase: a long period during which the triad behaves as three independent cores, with no early symmetry breaking.

Figure 6: Symmetric Triad Geometry (R = 30 px)



Initial configuration for the symmetric triad: three identical cores placed at the vertices of an equilateral triangle, all directed toward the centroid with equal speed v = 0.20. This geometry removes any privileged pair and tests whether ED supports a stable three-core configuration.

Figure 7: Symmetric Triad (400 Steps) — Symmetric Infall Phase



For the first 400 steps, the triad preserves symmetry to within numerical precision. All pairwise distances remain equal and far above the MERGE threshold. No interactions occur on this timescale, establishing the symmetric infall phase.

4.2 Symmetric Triad (1600 steps)

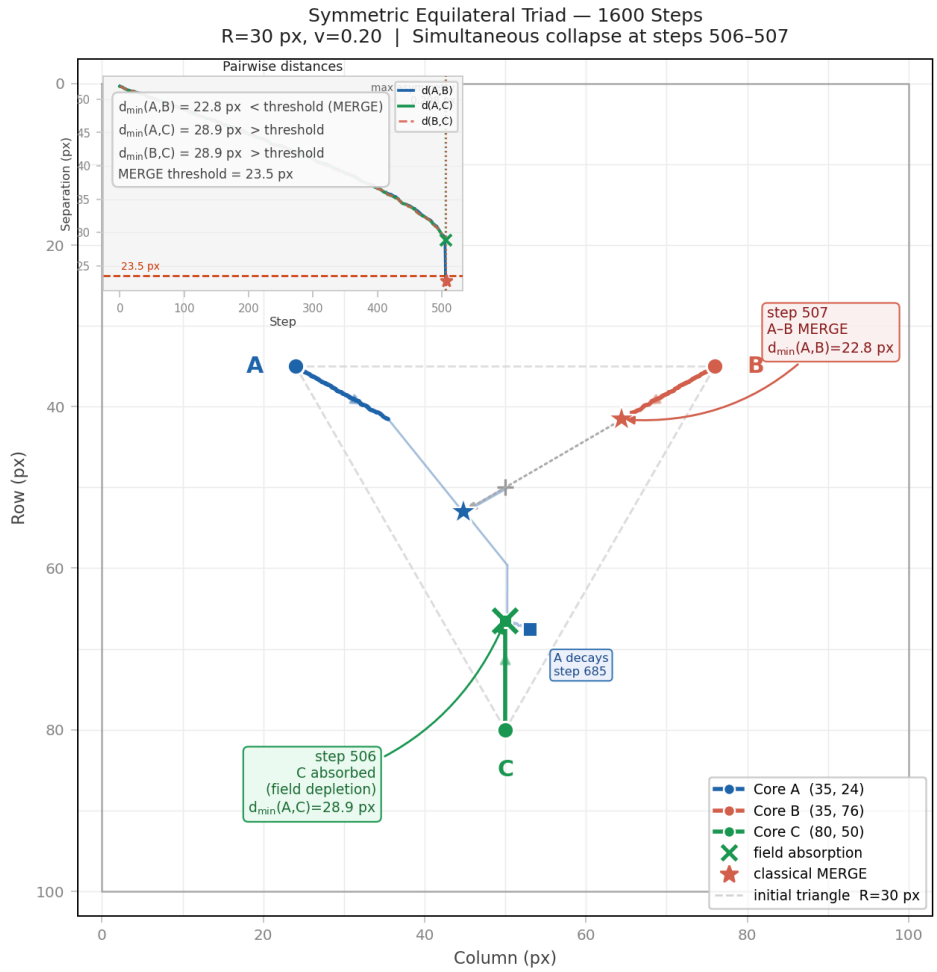
Extending the same configuration reveals the full triadic collapse. At step ~506, core C is absorbed by the shared field at a distance well above the MERGE threshold. One step later, cores A and B undergo a classical MERGE at $D_{\min}$ ≈22.8 px. The merged core then decays naturally.

The collapse is effectively 3 → 1, not 3 → 2 → 1. The triad does not pass through a stable two-core intermediate state. Instead, the system undergoes a burst collapse: one core is absorbed non-kinematically, and the remaining pair merges immediately afterward.

The key architectural result is the spontaneous symmetry breaking. Although the initial geometry is perfectly equilateral, the A–B pair is selected as the MERGE channel because their velocity components point directly toward each other along a horizontal axis. The oblique A–C and B–C approaches never reach the same depth of interaction. ED amplifies this geometric asymmetry and collapses the triad along a single privileged axis.

This experiment demonstrates that ED does not support a stable three-core configuration at these parameters. Symmetry is preserved during infall but broken decisively at collapse.

Figure 8: Symmetric Triad Collapse (1600 Steps) — Spontaneous Symmetry Breaking



At step ~506, core C is absorbed by the shared field at above-threshold distance. One step later, A and B undergo a classical MERGE. The collapse is effectively 3 → 1, not 3 → 2 → 1. The system selects the A–B axis as the collapse channel, demonstrating spontaneous symmetry breaking in a perfectly symmetric configuration.

5. Experiment Series 3 — R × v Sweep for Long-Lived Triads

The final experimental series tests whether any combination of initial separation *R* and inward speed *v* can produce a long-lived three-core configuration. A configuration is considered long-lived if all three cores remain distinct for at least 2000 steps.

Two circumradii (R = 30, 40) are physically valid on the current domain. Larger values (R = 50, 60) produce wrapped or off-grid geometries and are excluded from physical interpretation.

5.1 Results for R = 30

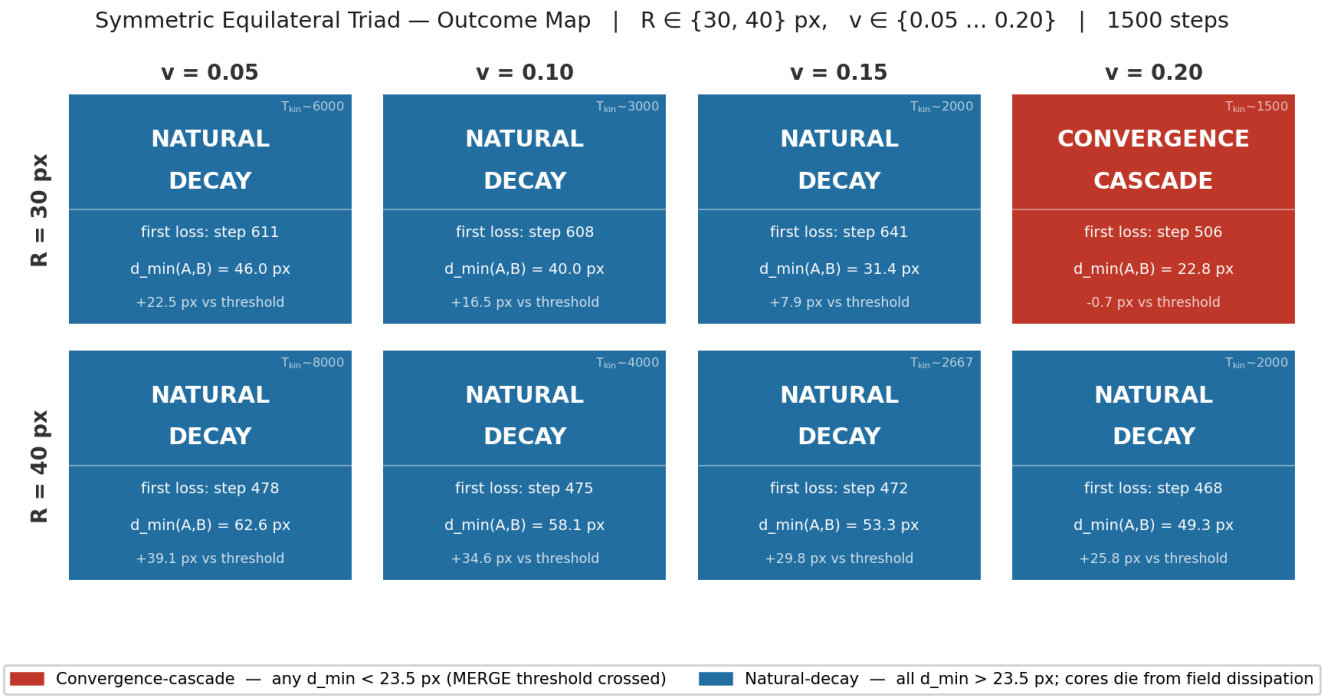
For *v*=0.05,0.10,0.15, the bottom vertex C decays naturally at ~600–640 steps. At the moment of decay, all pairwise separations remain far above the MERGE threshold (31–46 px). No interactions occur. The triad dissolves via single-core lifetime, not via triadic dynamics.

At *v* = 0.20, convergence is fast enough that A–C and B–C reach the MERGE threshold around step ~500. This triggers a symmetry-breaking cascade similar to Experiment 2b. The triad collapses rapidly and does not form a long-lived structure.

5.2 Results for R = 40

For all speeds $v = 0.05 – 0.20$, C decays at ~470–480 steps. Pairwise separations remain large (49–62 px). No pair approaches the MERGE threshold. The triad evaporates before any interaction can develop.

Figure 9: R × v Sweep — Natural-Decay vs Convergence-Cascade Regimes



Outcome grid for symmetric triads across circumradius R and inward speed v. For R = 30 and $v \leq 0.15$, cores decay individually before interacting. For R = 30 and v = 0.20, convergence is fast enough to trigger a collapse. For R = 40, all speeds lie in the natural-decay regime. No (R, v) combination yields a long-lived three-core configuration.

5.3 Phase Boundary

A clear transition emerges:

- Natural-decay regime:
  Large R or small v → cores approach too slowly to interact before their intrinsic lifetime expires.

- Convergence-cascade regime:
  Small R and large v → cores reach the MERGE threshold before decay, triggering a rapid collapse.

No region of the (R, v) plane supports a stable or even metastable three-core configuration.

5.4 Architectural Implication

The limiting factor is not geometry but lifetime. With $\alpha = 0.001$, a single core lives ~634 steps. Unless convergence is extremely fast, the triad decays before interacting. When convergence is fast enough, the triad collapses immediately. In neither case does a three-core molecule emerge.

6. The Triadic Interaction Laws

The full experimental arc reveals three architectural laws governing triadic behavior in ED. These laws generalize the two-core interaction law and mark the first departure from pairwise compositionality.

Triadic Law 1 — Blocking and Field Absorption

A bound pair generates a shared field that suppresses bilateral MERGE with a third core. The intruder is absorbed by the composite field before any two-core interaction can form. This mechanism has no analogue in the bilateral system.

Triadic Law 2 — Mechanism-Flip Boundary

There exists a structured frontier separating four regimes:

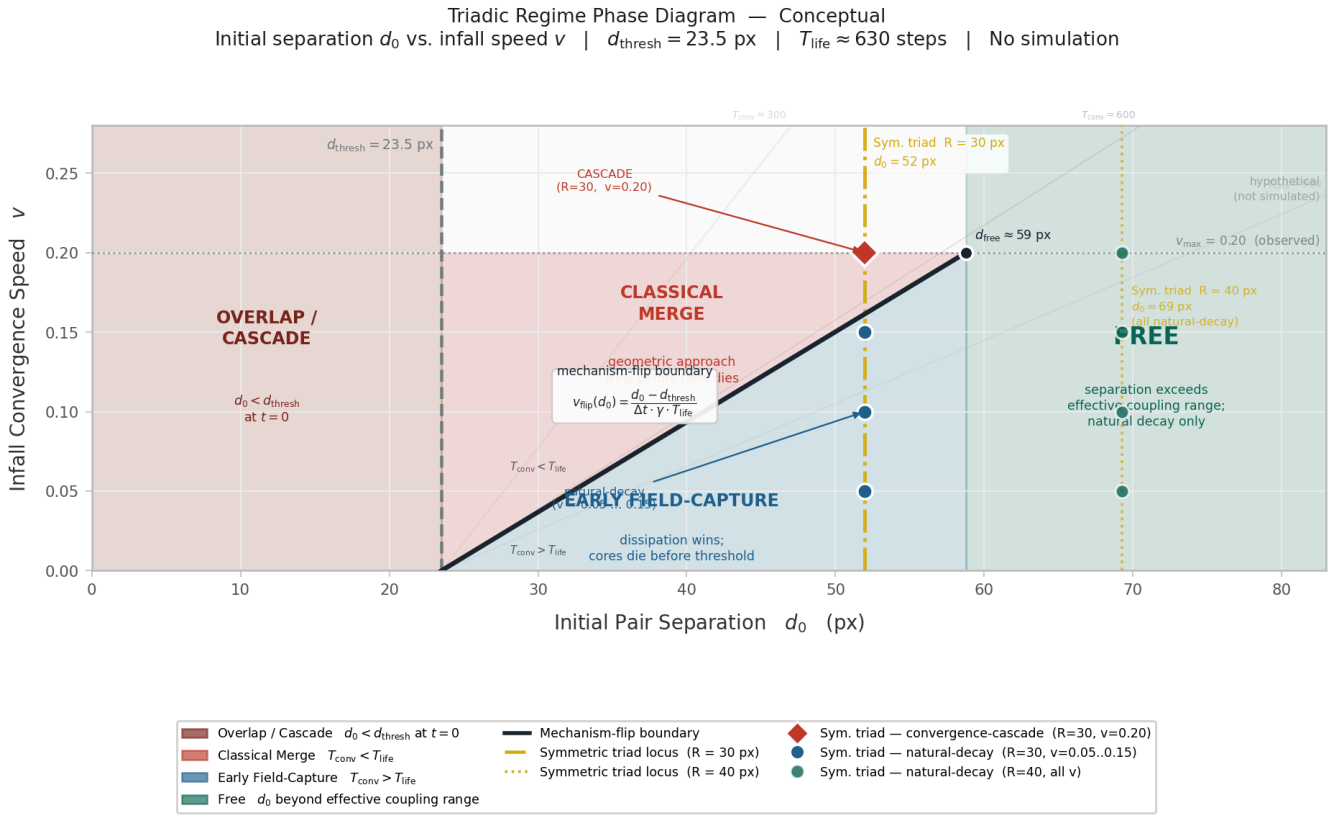
1. free motion,
2. classical MERGE,
3. early field-capture,
4. overlap/cascade collapse.

The transition between classical MERGE and early field-capture is a mechanism flip, not a smooth threshold. This boundary defines the first triadic phase diagram in ED.

Triadic Law 3 — Symmetry Breaking in Symmetric Triads

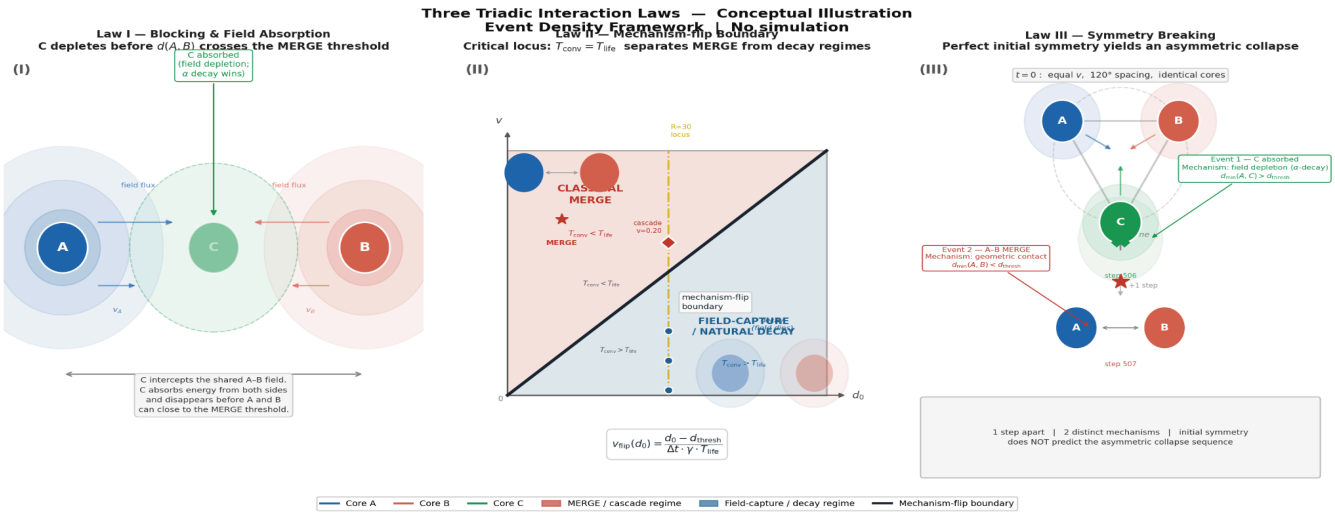
Even perfectly symmetric triads collapse by selecting a privileged pair. The collapse axis is determined by geometric alignment of velocity components, not by initial distances. ED does not support a stable three-core bound state at the tested parameters.

Figure 10: Triadic Phase Diagram



Schematic phase diagram summarizing the four triadic regimes: free motion, classical MERGE, early field-capture, and overlap/cascade collapse. The mechanism-flip boundary between classical MERGE and early capture is highlighted. Symmetric triads trace a single line through this diagram and never enter a stable region.

Figure 11: The Triadic Interaction Laws



Conceptual illustration of the three Triadic Interaction Laws: (1) blocking and field absorption, (2) the mechanism-flip boundary, and (3) symmetry breaking in symmetric triads. These laws formalize the architectural behavior of three-core systems in ED and demonstrate the breakdown of pairwise compositionality.

7. Triadic Phase Diagram

The full experimental arc reveals that triadic behavior in ED is not a simple extension of the two-core law. Instead, it organizes into a multi-regime phase structure defined by the interplay of distance, velocity, and shared-field depth. The phase diagram is constructed from the intruder sweep, the symmetric triad collapse, and the R×v parameter study.

7.1 Axes of the Diagram

- Horizontal axis: initial intruder distance (or effective separation in symmetric triads).
- Vertical axis: kinematic convergence rate (controlled by v and geometry).
- Implicit axis: field depth (determined by α and amplitude, held fixed in this paper).

7.2 The Four Regimes

Across these axes, triadic dynamics fall into four reproducible regimes:

- Free Regime
  Cores remain too distant for interaction within the available lifetime.
  Pairwise distances stay far above the MERGE threshold.

- Classical MERGE Regime
  A bilateral MERGE occurs exactly as predicted by the two-core law.
  This regime survives only in narrow geometric corridors.

- Early Field-Capture Regime
  A third core is absorbed by the shared field of a pair before any bilateral MERGE can form.
  This mechanism has no two-core analogue and marks the breakdown of pairwise compositionality.

- Overlap / Cascade Regime
  Cores begin inside each other's Gaussian support or converge so rapidly that collapse occurs in a burst.
  The system transitions directly from 3 → 1 without a stable two-core intermediate.

7.3 Symmetric Triads as a Special Line

Symmetric triads trace a single line through this diagram:

- long symmetric infall,
- sudden symmetry breaking,
- collapse along a privileged axis.

They never enter a stable region. No basin of attraction exists for a three-core bound state at the tested parameters.

7.4 Architectural Interpretation

The triadic phase diagram shows that ED's many-body behavior is structured but not compositional. The two-core law governs only a subset of triadic interactions; new mechanisms dominate once a third core is present. This is the central architectural result of ED-Arch-13.

8. Architectural Consequences

The triadic experiments reveal a decisive architectural fact: pairwise laws do not scale. ED's many-body behavior is governed by field-level interactions that cannot be reduced to bilateral MERGE/BIND/SCATTER rules.

8.1 Failure of Pairwise Composition

The two-core law predicts that A–C should MERGE in many intruder geometries. Instead:

- C is absorbed early by the A–B shared field, or
- collapse occurs through a triadic cascade.

The bilateral law is not violated — it is overridden by triadic field structure.

8.2 No Stable Three-Core Structures

Across all physically valid (R, v) combinations:

- triads either decay before interacting, or
- collapse rapidly once they do.

The absence of a stable three-core configuration is not an accident of geometry; it is a structural feature of ED at $\alpha = 0.001$.

8.3 Symmetry Breaking as an Architectural Principle

Even perfectly symmetric triads collapse asymmetrically.
The collapse axis is determined by velocity alignment, not by initial distances.
This is a new architectural invariant: triads select a privileged pair.

8.4 Implications for Many-Body ED

The triadic laws imply that:

- ED's many-body dynamics require new laws beyond the two-core framework.
- Higher-order interactions are not reducible to pairwise terms.
- The shared field acts as a composite object with its own interaction rules.

These insights set the stage for ED-Arch-14, where dissipation and amplitude will be tuned to probe whether triadic stability can ever emerge.

9. Conclusion

ED-Arch-13 establishes the first complete picture of triadic interactions in Event Density. Through a sequence of controlled experiments — intruder triads, symmetric triads, and a systematic R×v sweep — we find that the two-core interaction law does not compose. Instead, triads exhibit new mechanisms: blocking, early field-capture, overlap collapse, and spontaneous symmetry breaking.

From these results, we derive the Triadic Interaction Laws, which govern how three cores behave in ED. These laws define a structured phase diagram with four regimes and show that no stable three-core configuration exists at the tested parameters.

The architectural consequence is clear: ED's many-body behavior is not pairwise-compositional. Triads introduce qualitatively new dynamics that require their own laws. ED-Arch-14 will extend this investigation by tuning α and

amplitude to search for self-sustaining triads and by introducing controlled asymmetries to probe how collapse channels can be biased.

ED-Arch-13 closes the bilateral era. The architecture now moves into the genuinely many-body domain.